

DIFFERENTIATING INVERSE TRIG FUNCTIONS

A short set of notes for differentiating the inverse trig functions.

Sine, cosine, and tangent, together with their reciprocals cosecant, secant, and cotangent, form the basic trigonometric functions. Recall that

$$\frac{d}{dx} \sin(x) = \cos(x), \quad \frac{d}{dx} \cos(x) = -\sin(x), \quad \frac{d}{dx} \tan(x) = \sec^2(x),$$

where $\csc(x) = 1/\sin(x)$, $\sec(x) = 1/\cos(x)$, and $\cot(x) = 1/\tan(x)$. In these notes we derive the derivatives for $\csc(x)$, $\sec(x)$, and $\cot(x)$.

All three can be found through a combination of the chain, product, or quotient rule.

Chain and product rule approach.

- (a) Let $y = \csc(x) = 1/\sin(x) = [\sin(x)]^{-1}$. We know u^{-1} differentiates to $-u^{-2}$. Adjusting with the derivative of $\sin(x)$, we have

$$\frac{dy}{dx} = \cos(x) \cdot -\frac{1}{\sin^2(x)} = -\frac{\cos(x)}{\sin^2(x)} = -\frac{1}{\sin(x)} \frac{\cos(x)}{\sin(x)} = -\csc(x) \cot(x).$$

So $\csc(x)$ differentiates to $-\csc(x) \cot(x)$.

- (b) Let $y = \sec(x) = 1/\cos(x) = [\cos(x)]^{-1}$. Again, we know u^{-1} differentiates to $-u^{-2}$. Adjusting with the derivative of $\cos(x)$, we have

$$\frac{dy}{dx} = -\sin(x) \cdot -\frac{1}{\cos^2(x)} = \frac{\sin(x)}{\cos^2(x)} = \frac{1}{\cos(x)} \frac{\sin(x)}{\cos(x)} = \sec(x) \tan(x).$$

So $\sec(x)$ differentiates to $\sec(x) \tan(x)$.

- (c) Let $y = \cot(x) = \cos(x)/\sin(x)$. Write this as $\cos(x) \cdot 1/\sin(x) = \cos(x) \csc(x)$ and use the product rule. To be clear,

$$y = \cot(x) = \cos(x) \csc(x) =: u \cdot v,$$

i.e. $u = \cos(x)$ and $v = \csc(x)$. We know $u' = -\sin(x)$ and $v' = -\csc(x) \cot(x)$. Then

$$\begin{aligned} \frac{dy}{dx} &= vu' + uv' \\ &= -\csc(x) \sin(x) - \cos(x) \csc(x) \cot(x) \\ &= -\frac{1}{\sin(x)} \sin(x) - \cos(x) \frac{1}{\sin(x)} \frac{\cos(x)}{\sin(x)} \\ &= -1 - \frac{\cos^2(x)}{\sin^2(x)} = -(1 + \cot^2(x)) = -\csc^2(x) \end{aligned}$$

using the identity $1 + \cot^2(x) = \csc^2(x)$. So $\cot(x)$ differentiates to $-\csc^2(x)$.

It is also possible to use the quotient rule to achieve the same results. Recall this rule says that if $y = \frac{u}{v}$ then

$$\frac{dy}{dx} = \frac{vu' - uv'}{v^2}.$$

Quotient rule approach.

- (a) Let $y = \csc(x) = 1/\sin(x)$. Then $u = 1, u' = 0$ and $v = \sin(x), v' = \cos(x)$, so

$$\frac{dy}{dx} = \frac{-1 \cdot \cos(x)}{\sin^2(x)} = -\frac{1}{\sin(x)} \frac{\cos(x)}{\sin(x)} = -\csc(x) \cot(x).$$

Again, $\csc(x)$ differentiates to $-\csc(x) \cot(x)$.

- (b) Let $y = \sec(x) = 1/\cos(x)$. Then $u = 1, u' = 0$ and $v = \cos(x), v' = -\sin(x)$, so

$$\frac{dy}{dx} = \frac{-1 \cdot -\sin(x)}{\cos^2(x)} = \frac{\sin(x)}{\cos^2(x)} = \frac{1}{\cos(x)} \frac{\sin(x)}{\cos(x)} = \sec(x) \tan(x).$$

Again, $\sec(x)$ differentiates to $\sec(x) \tan(x)$.

- (c) Let $y = \cot(x) = \cos(x)/\sin(x)$. Then $u = \cos(x), u' = -\sin(x)$ and $v = \sin(x), v' = \cos(x)$, so

$$\begin{aligned} \frac{dy}{dx} &= \frac{\sin(x) \cdot -\sin(x) - \cos(x) \cdot \cos(x)}{\sin^2(x)} \\ &= \frac{-\sin^2(x) - \cos^2(x)}{\sin^2(x)} = -1 - \frac{\cos^2(x)}{\sin^2(x)} \\ &= -(1 + \cot^2(x)) = -\csc^2(x) \end{aligned}$$

using the identity $1 + \cot^2(x) = \csc^2(x)$. Again, $\cot(x)$ differentiates to $-\csc^2(x)$.